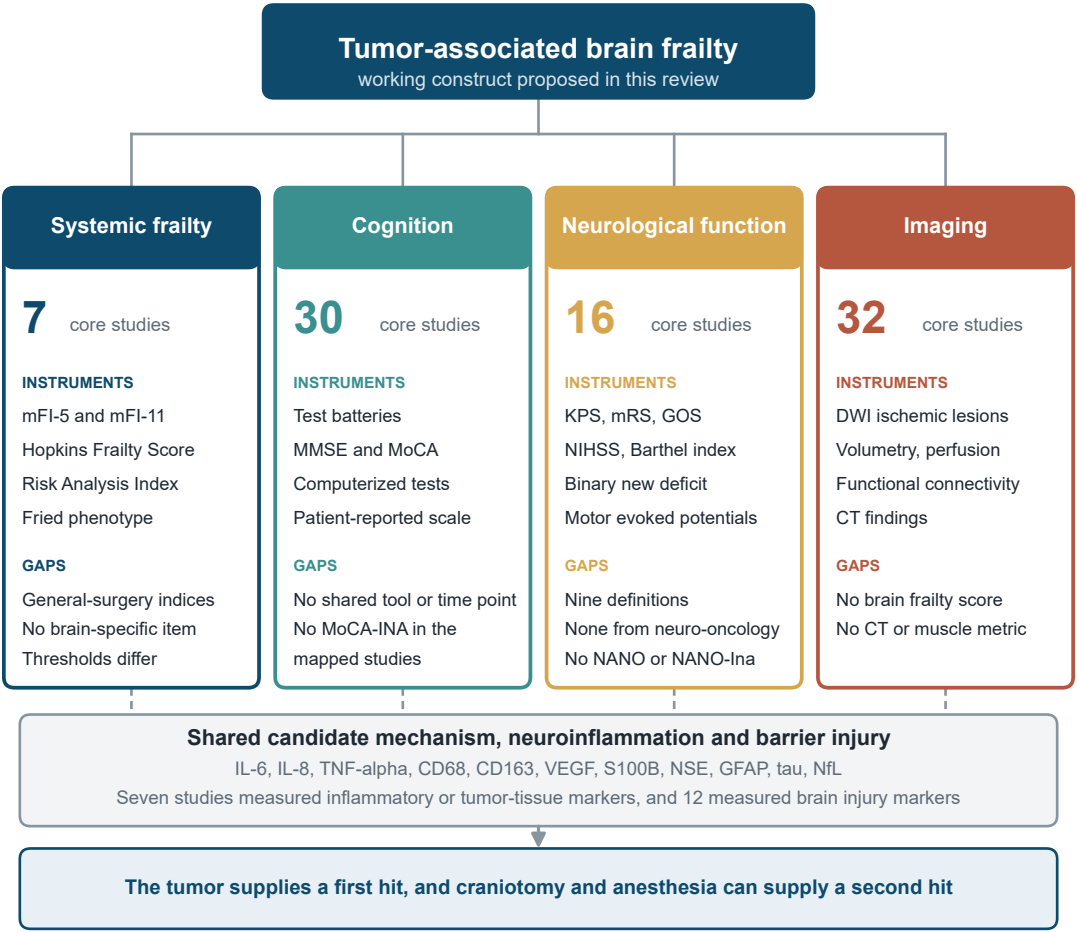

Tumor-associated brain frailty
working construct proposed in this review
Systemic frailty
7 core studies
INSTRUMENTS
mFI-5 and mFI-11
Hopkins Frailty Score
Risk Analysis Index
Fried phenotype
GAPS
General-surgery indices
No brain-specific item
Thresholds differ
Cognition
30 core studies
INSTRUMENTS
Test batteries
MMSE and MoCA
Computerized tests
Patient-reported scale
GAPS
No shared tool or time point
No MoCA-INA in the
mapped studies
Neurological function
16 core studies
INSTRUMENTS
KPS, mRS, GOS
NIHSS, Barthel index
Binary new deficit
Motor evoked potentials
GAPS
Nine definitions
None from neuro-oncology
No NANO or NANO-Ina
Imaging
32 core studies
INSTRUMENTS
DWI ischemic lesions
Volumetry, perfusion
Functional connectivity
CT findings
GAPS
No brain frailty score
No CT or muscle metric
Shared candidate mechanism, neuroinflammation and barrier injury
IL-6, IL-8, TNF-alpha, CD68, CD163, VEGF, S100B, NSE, GFAP, tau, NfL
Seven studies measured inflammatory or tumor-tissue markers, and 12 measured brain injury markers
The tumor supplies a first hit, and craniotomy and anesthesia can supply a second hit